**Institut Teknologi Bandung**

Directorate of Continuing Professional Education

in partnership with

CHAPTER 7

# A Deep Dive on Keras

Three ways to build a model, three levels of control over training — and one principle holding them together: easy to start, and no ceiling.

**Duration** 3 hours (2 sessions) **Instructor** Rahman Indra Kesuma, S.Kom., M.Cs.

**Source** Chollet & Watson, Deep Learning with Python 3e — chapter 7

LEARNING OUTCOMES

# Session Objectives

By the end of this session, participants can:

1. Explain **progressive disclosure of complexity**, and place yourself on the spectrum of Keras workflows.
2. Build multi-input, multi-output models with the **Functional API**, and train them with either lists or dictionaries.
3. Use **access to layer connectivity**: plot a topology with `plot_model()` and **extract features** from intermediate nodes.
4. Write a **Model subclass** — and name what you give up by choosing it.
5. Write **custom metrics** (`update_state`, `result`, `reset_state`) and **custom callbacks**, and combine `EarlyStopping` with `ModelCheckpoint`.
6. Write your own **`train_step()`** in TensorFlow, PyTorch, and JAX, and plug it into `fit()` so callbacks and built-in optimisations still apply.

BOOK
Chapter 7 — full text

NOTEBOOK
01_three_ways_to_build_a_model.ipynb

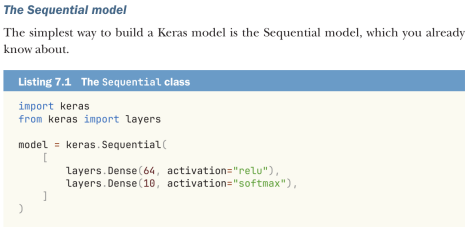
NOTEBOOK
02_custom_metrics_and_callbacks.ipynb

NOTEBOOK
03_custom_train_step_per_backend.ipynb

REPO
Author's official notebook

SECTION 7.1

# One principle behind the whole API



***The Sequential model***

The simplest way to build a Keras model is the Sequential model, which you already know about.

**Listing 7.1 The `Sequential` class**

```
import keras
from keras import layers

model = keras.Sequential(
    [
        layers.Dense(64, activation="relu"),
        layers.Dense(10, activation="softmax"),
    ]
)
```

Figure 7.1 — not four different frameworks, but one spectrum built on shared APIs. — Chollet & Watson, *Deep Learning with Python*, 3rd ed. (Manning)

> *You could be using Keras like you would use scikit-learn — just calling fit() and letting the framework do its thing — or you could be using it like NumPy, taking full control of every little detail.*
>
> Chollet & Watson, section 7.1

SECTION 7.1

# Why that matters for a career, not just a project

> So what you learn today stays valid once you are an expert. You do **not** have to switch frameworks going from student to researcher, or from data scientist to deep learning engineer.

The book's analogy: **Keras is the Python of deep learning** — multiparadigm, with several usage patterns that work together rather than one "true" way.

01

# Three ways to build a model

Sequential, Functional, subclassing — and when each is right.

SECTION 7.2.1

# Sequential is essentially a Python list

In practice, models with **multiple inputs** (an image and its metadata), **multiple outputs** (several things to predict), or a **non-linear topology** are **entirely ordinary**

SECTION 7.2.2

# The Functional API, and the symbolic tensor

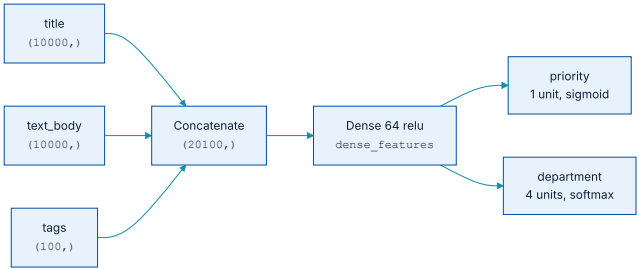
listing 7.8 — a simple Functional model PYTHON

```python
inputs = keras.Input(shape=(3,), name="my_input")
features = layers.Dense(64, activation="relu")(inputs)
outputs = layers.Dense(10, activation="softmax")(features)

model = keras.Model(inputs=inputs, outputs=outputs, name="my_functional_model")

print(inputs.shape, inputs.dtype)      # (None, 3) float32  -- None = batch size
print(features.shape)                 # (None, 64)
```

`inputs` is a **symbolic tensor**: it holds **no data at all**, only the specification of the tensors the model will eventually see. Every Keras layer accepts either real tensors or symbolic ones.

SECTION 7.2.2

# A model that Sequential cannot express




Three inputs, one shared intermediate representation, two heads.

SECTION 7.2.2 · LISTING 7.9

# . . . and it is nine lines

listing 7.9 — multi-input, multi-output PYTHON

```python
vocabulary_size, num_tags, num_departments = 10000, 100, 4

title     = keras.Input(shape=(vocabulary_size,), name="title")
text_body = keras.Input(shape=(vocabulary_size,), name="text_body")
tags      = keras.Input(shape=(num_tags,), name="tags")

features = layers.Concatenate()([title, text_body, tags])
features = layers.Dense(64, activation="relu", name="dense_features")(features)

priority   = layers.Dense(1, activation="sigmoid", name="priority")(features)
department = layers.Dense(num_departments, activation="softmax",
                          name="department")(features)

model = keras.Model(inputs=[title, text_body, tags],
                    outputs=[priority, department])
```

The book describes it as **like playing with LEGO bricks**: simple, and flexible enough for arbitrary graphs of layers.

# Training it: by position, or by name

listing 7.10 — lists PYTHON

```python
model.compile(
    optimizer="adam",
    loss=["mean_squared_error",
          "sparse_categorical_crossentropy"],
    metrics=[["mean_absolute_error"],
             ["accuracy"]])

model.fit(
    [title_data, text_body_data, tags_data],
    [priority_data, department_data],
    epochs=1)
```

listing 7.11 — dictionaries PYTHON

```python
model.compile(
    optimizer="adam",
    loss={"priority": "mean_squared_error",
          "department":
              "sparse_categorical_crossentropy"},
    metrics={"priority": ["mean_absolute_error"],
             "department": ["accuracy"]})

model.fit(
    {"title": title_data,
     "text_body": text_body_data,
     "tags": tags_data},
    {"priority": priority_data,
     "department": department_data},
    epochs=1)
```

The list form **must follow the order** given to `Model()`. With many inputs or outputs, **use the dictionary form** — order stops mattering and the code becomes far harder to get wrong.

# The real payoff: the graph is inspectable

plotting and inspecting PYTHON

```python
keras.utils.plot_model(model, "ticket_classifier.png")

# far more useful while debugging:
keras.utils.plot_model(model, "ticket_classifier_with_shape_info.png",
                       show_shapes=True, show_layer_names=True)

print(model.layers[3].output)
```

OUTPUT

```
<KerasTensor shape=(None, 20100), dtype=float32>
```

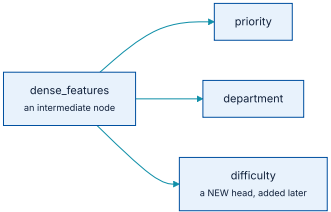
The `None` is the batch size: this model accepts batches of any size.

# Feature extraction: adding a head without retraining

listing 7.13 — reusing an intermediate node

PYTHON

```
features = model.layers[4].output          # the shared Dense layer
difficulty = layers.Dense(3, activation="softmax", name="difficulty")(features)

new_model = keras.Model(
    inputs=[title, text_body, tags],
    outputs=[priority, department, difficulty])
```




One shared representation, now feeding three heads.

# Model subclassing: total control

listing 7.14 — the same model, subclassed PYTHON

```python
class CustomerTicketModel(keras.Model):
    def __init__(self, num_departments):
        super().__init__()                          # do not forget the super constructor
        self.concat_layer = layers.Concatenate()
        self.mixing_layer = layers.Dense(64, activation="relu")
        self.priority_scorer = layers.Dense(1, activation="sigmoid")
        self.department_classifier = layers.Dense(num_departments,
                                                  activation="softmax")

    def call(self, inputs):                         # the forward pass lives here
        features = self.concat_layer(
            [inputs["title"], inputs["text_body"], inputs["tags"]])
        features = self.mixing_layer(features)
        return self.priority_scorer(features), self.department_classifier(features)
```

This unlocks models that **cannot be expressed as a directed acyclic graph** — a `call()` that uses layers inside a `for` loop, or calls them recursively.

SECTION 7.2.3

# Layer or Model — what actually differs

| | `Layer` | `Model` |
|---|---|---|
| Role | A building block used to make models. | The top-level object you train and export. |
| `fit()`, `evaluate()`, `predict()` | Does not have them. | Has them. |
| Saveable to a file | No. | Yes. |

Beyond that the two classes are **virtually identical**.

SECTION 7.2.3

# What subclassing costs you

**Lost when you subclass**

`summary()` does not show layer connectivity · `plot_model()` cannot be used · feature extraction is impossible — there is simply no graph.

**A Functional model** is an explicit data structure — a graph of layers you can view, inspect, and modify.

**A subclassed model** is **a piece of bytecode**: a Python class with a `call()` method containing raw code. That is the source of its flexibility, and of its limitations.

Once instantiated, the forward pass becomes **a complete black box**. You are developing a new Python object rather than snapping bricks together, so **the error surface is much larger**

# The three mix freely

listing 7.15 — subclass inside Functional PYTHON

```python
class Classifier(keras.Model):
    def __init__(self, num_classes=2):
        super().__init__()
        units, act = ((1, "sigmoid") if num_classes == 2
                      else (num_classes, "softmax"))
        self.dense = layers.Dense(units, activation=act)

    def call(self, inputs):
        return self.dense(inputs)

inputs = keras.Input(shape=(3,))
features = layers.Dense(64, activation="relu")(inputs)
outputs = Classifier(num_classes=10)(features)
model = keras.Model(inputs, outputs)
```

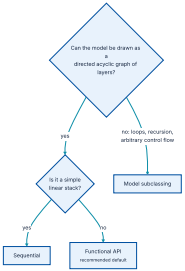
listing 7.16 — Functional inside subclass PYTHON

```python
inputs = keras.Input(shape=(64,))
outputs = layers.Dense(1, activation="sigmoid")(inputs)
binary_classifier = keras.Model(inputs, outputs)

class MyModel(keras.Model):
    def __init__(self):
        super().__init__()
        self.dense = layers.Dense(64, activation="relu")
        self.classifier = binary_classifier

    def call(self, inputs):
        return self.classifier(self.dense(inputs))
```

They are all part of the same spectrum, so the boundary is a design choice rather than a wall.

SECTION 7.2.5

# Which one to reach for




The book's own recommendation, as a decision.

Every example in the rest of the book uses the **Functional API** — but with **subclassed layers** inside it. That combination gives **development flexibility while keeping the Functional API's advantages**

02

# The built-in training loop

Custom metrics, callbacks, and TensorBoard.

SECTION 7.3

# The workflow you already know

listing 7.17 — the standard workflow PYTHON

```python
def get_mnist_model():
    inputs = keras.Input(shape=(28 * 28,))
    features = layers.Dense(512, activation="relu")(inputs)
    features = layers.Dropout(0.5)(features)
    outputs = layers.Dense(10, activation="softmax")(features)
    return keras.Model(inputs, outputs)

model = get_mnist_model()
model.compile(optimizer="adam", loss="sparse_categorical_crossentropy",
              metrics=["accuracy"])
model.fit(train_images, train_labels, epochs=3,
          validation_data=(val_images, val_labels))
test_metrics = model.evaluate(test_images, test_labels)
predictions = model.predict(test_images)
```

Two ways to customise it: **your own metrics**, and **callbacks** scheduled at specific points during training.

# Writing your own metric

listing 7.18 — RMSE as a custom metric

PYTHON

```python
from keras import ops

class RootMeanSquaredError(keras.metrics.Metric):
    def __init__(self, name="rmse", **kwargs):
        super().__init__(name=name, **kwargs)
        self.mse_sum = self.add_weight(name="mse_sum", initializer="zeros")
        self.total_samples = self.add_weight(name="total_samples", initializer="zeros")

    def update_state(self, y_true, y_pred, sample_weight=None):
        y_true = ops.one_hot(y_true, num_classes=ops.shape(y_pred)[1])
        self.mse_sum.assign_add(ops.sum(ops.square(y_true - y_pred)))
        self.total_samples.assign_add(ops.shape(y_pred)[0])

    def result(self):
        return ops.sqrt(self.mse_sum / self.total_samples)

    def reset_state(self):
        self.mse_sum.assign(0.)
        self.total_samples.assign(0.)
```

The next slide names what each of those three methods is responsible for.

SECTION 7.3.1

# The three methods that form the contract

**update_state()**
Called once per batch with the targets and predictions. This is where the accumulation happens.

**result()**
Returns the metric's current value from that accumulated state.

**reset_state()**
Clears the state without rebuilding the object, so the same instance serves every epoch and both training and evaluation.

Unlike a layer, none of this state is touched by backpropagation — **you own the update logic entirely**

SECTION 7.3.1

# ...and using it is the same as any built-in

using the custom metric

PYTHON

```python
model = get_mnist_model()
model.compile(optimizer="adam",
              loss="sparse_categorical_crossentropy",
              metrics=["accuracy", RootMeanSquaredError()])
model.fit(train_images, train_labels, epochs=3,
          validation_data=(val_images, val_labels))
```

OUTPUT

```
accuracy: 0.9612 - loss: 0.1284 - rmse: 0.2371
```

SECTION 7.3.2

# Callbacks: from a paper aeroplane to a drone

> *Launching a training run on a large dataset for tens of epochs using model.fit() can be a bit like launching a paper airplane: past the initial impulse, you don't have any control over its trajectory or its landing spot.*
>
> Chollet & Watson, section 7.3.2

A callback is an object passed to `fit()` that Keras calls at various points during training. It can see the model's state and **act**: interrupt training, save the model, load different weights.

SECTION 7.3.2

# What callbacks are used for

**Checkpointing**
Saving the model state at different points during training.

**Early stopping**
Halting when validation stops improving — and keeping the best model found.

**Dynamic parameters**
Adjusting things like the optimizer's learning rate mid-run.

**Logging and plotting**
The `fit()` progress bar you already know is **itself a callback**

some of the built-in callbacks — PYTHON

```
keras.callbacks.ModelCheckpoint
keras.callbacks.EarlyStopping
keras.callbacks.LearningRateScheduler
keras.callbacks.ReduceLROnPlateau
keras.callbacks.CSVLogger
```

Two of these do most of the work in practice, and the next slide shows them used together.

# The standard pair

listing 7.19 — callbacks in fit() PYTHON

```python
callbacks_list = [
    keras.callbacks.EarlyStopping(
        monitor="accuracy",     # must be among the model's metrics
        patience=1,             # stop if it has not improved for one epoch
    ),
    keras.callbacks.ModelCheckpoint(
        filepath="checkpoint_path.keras",
        monitor="val_loss",
        save_best_only=True,    # only overwrite when val_loss improves
    ),
]

model.fit(train_images, train_labels, epochs=10,
          callbacks=callbacks_list,
          validation_data=(val_images, val_labels))   # REQUIRED: val_* is monitored
```

This replaces the wasteful pattern from chapters 4 and 5 — train to find the best epoch, then retrain from scratch. **One run is now enough.**

SECTION 7.3.2

# Saving and reloading by hand

manual save and load

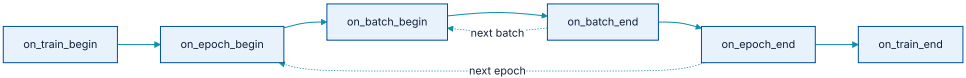
PYTHON

```python
model.save("my_checkpoint_path.keras")
model = keras.models.load_model("checkpoint_path.keras")
```

This is the format chapter 6 exports from when preparing a model for deployment.

SECTION 7.3.3

# Six hooks you can implement




Each hook receives a `logs` dictionary with the metrics available at that moment.

# A custom callback, end to end

listing 7.20 — per-batch loss history PYTHON

```python
from matplotlib import pyplot as plt

class LossHistory(keras.callbacks.Callback):
    def on_train_begin(self, logs):
        self.per_batch_losses = []

    def on_batch_end(self, batch, logs):
        self.per_batch_losses.append(logs.get("loss"))

    def on_epoch_end(self, epoch, logs):
        plt.clf()
        plt.plot(range(len(self.per_batch_losses)), self.per_batch_losses,
                 label="Training loss for each batch")
        plt.xlabel(f"Batch (epoch {epoch})")
        plt.ylabel("Loss")
        plt.savefig(f"plot_at_epoch_{epoch}", dpi=300)
        self.per_batch_losses = []
```

Pass it as `callbacks=[LossHistory()]`. Useful when a run looks unstable and you need to see **whether the instability is within an epoch or between them**

SECTION 7.3.4

# TensorBoard closes the experiment loop

**Monitor metrics**
Visually, while training runs.

**Visualise the architecture**
The model graph.

**Histograms**
Of activations and gradients.

**Explore embeddings**
In 3D.

SECTION 7.3.4

# Turning it on

the callback

PYTHON

```python
tensorboard = keras.callbacks.TensorBoard(log_dir="/full_path_to_your_log_dir")

model.fit(train_images, train_labels, epochs=10,
          validation_data=(val_images, val_labels),
          callbacks=[tensorboard])
```

and the viewer

BASH

```bash
# on a local machine
tensorboard --logdir /full_path_to_your_log_dir

# inside a Colab notebook
%load_ext tensorboard
%tensorboard --logdir /full_path_to_your_log_dir
```

If `tensorboard` is not on your PATH, it installs with `pip install tensorboard`.

# 03

# Writing your own training loop

When fit() is not enough — and how to keep it anyway.

SECTION 7.4

# Where the built-in loop stops

**Generative learning**
No explicit targets at all. Introduced in **chapter 16**.

**Self-supervised**
Targets are taken **from the inputs themselves**.

**Reinforcement learning**
Driven by occasional **rewards** — much like training a dog.

SECTION 7.4.1

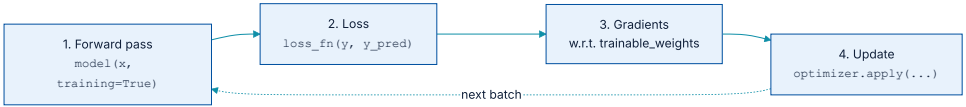
# Two details that break custom loops

1. **Pass `training=True` on the forward pass.** Layers such as `Dropout` behave differently during training and inference. `dropout(inputs, training=True)` drops activations; `training=False` does nothing. So: `predictions = model(inputs, training=True)`.
2. **Use `model.trainable_weights`, not `model.weights`.** Models hold two kinds: **trainable** weights updated by backpropagation, and **non-trainable** weights updated by the layers themselves during the forward pass.

> Among the built-in layers, the only one with non-trainable weights is `BatchNormalization` — it tracks the mean and standard deviation of the data flowing through it (chapter 9).

SECTION 7.4.2

# The shape of a training step




Identical in all three backends. Only step 3 is written differently.

SECTION 7.4.2

# TensorFlow and PyTorch, side by side

TensorFlow — PYTHON

```python
def train_step(inputs, targets):
    with tf.GradientTape() as tape:
        predictions = model(inputs, training=True)
        loss = loss_fn(targets, predictions)
    gradients = tape.gradient(loss, model.trainable_weights)
    optimizer.apply(gradients, model.trainable_weights)
    return loss
```

PyTorch — PYTHON

```python
def train_step(inputs, targets):
    predictions = model(inputs, training=True)
    loss = loss_fn(targets, predictions)
    loss.backward()                                # fill in the gradients
    gradients = [w.value.grad for w in model.trainable_weights]
    with torch.no_grad():                          # must be inside no_grad()
        optimizer.apply(gradients, model.trainable_weights)
    model.zero_grad()                              # backward() accumulates
    return loss
```

`model.zero_grad()` is **not optional** in PyTorch: `backward()` adds to the existing gradients, so without it the values pile up and **training will not proceed**

# JAX is harder, and the reason is statelessness

stateless_call and value_and_grad PYTHON

```python
outputs, non_trainable_weights = model.stateless_call(
    trainable_weights, non_trainable_weights, inputs)

def compute_loss_and_updates(trainable_variables, non_trainable_variables,
                             inputs, targets):
    outputs, non_trainable_variables = model.stateless_call(
        trainable_variables, non_trainable_variables, inputs, training=True)
    loss = loss_fn(targets, outputs)
    return loss, non_trainable_variables     # scalar FIRST, the rest is 'aux'

grad_fn = jax.value_and_grad(compute_loss_and_updates, has_aux=True)
(loss, non_trainable_weights), gradients = grad_fn(
    trainable_variables, non_trainable_variables, inputs, targets)
```

`has_aux=True` is required because `jax.grad` only accepts functions returning a scalar, and this one also returns the updated non-trainable weights.

# The optimizer has state too

stateless_apply PYTHON

```python
trainable_variables, optimizer_variables = optimizer.stateless_apply(
    optimizer_variables, gradients, trainable_variables)
```

The same pattern covers metrics: inside a stateless function you cannot call `update_state()`, so Keras provides **`stateless_update_state()`**, **`stateless_result()`**, and **`stateless_reset_state()`**

SECTION 7.4.3

# Using metrics at the low level

a normal metric — PYTHON

```python
metric = keras.metrics.SparseCategoricalAccuracy()
targets = ops.array([0, 1, 2])
predictions = ops.array([[1, 0, 0],
                         [0, 1, 0],
                         [0, 0, 1]])
metric.update_state(targets, predictions)
print(f"result: {metric.result():.2f}")
```

OUTPUT

```
result: 1.00
```

tracking a scalar average — PYTHON

```python
values = ops.array([0, 1, 2, 3, 4])
mean_tracker = keras.metrics.Mean()
for value in values:
    mean_tracker.update_state(value)
print(f"Mean: {mean_tracker.result():.2f}")
```

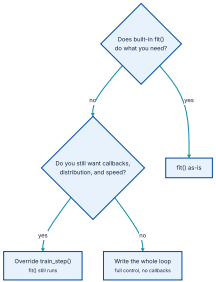
OUTPUT

```
Mean: 2.00
```

Remember `metric.reset_state()` **at the start of each training epoch and at the start of evaluation.** Forgetting it **mixes numbers across epochs** and quietly misleads you.

SECTION 7.4.4

# The middle ground




Writing the whole loop costs you callbacks, performance optimisations, and distributed training support.

# Override train_step(), keep fit()

listing 7.21 — a custom train_step PYTHON

```python
loss_fn = keras.losses.SparseCategoricalCrossentropy()
loss_tracker = keras.metrics.Mean(name="loss")

class CustomModel(keras.Model):
    def train_step(self, data):
        inputs, targets = data
        with tf.GradientTape() as tape:
            predictions = self(inputs, training=True)   # self, not model
            loss = loss_fn(targets, predictions)
        gradients = tape.gradient(loss, self.trainable_weights)
        self.optimizer.apply(gradients, self.trainable_weights)
        loss_tracker.update_state(loss)
        return {"loss": loss_tracker.result()}

    @property
    def metrics(self):
        return [loss_tracker]      # listed here so reset_state() is called for you
```

- Works whether you build with **Sequential, Functional, or subclassing**.
- You do **not** need `@tf.function` or `@jax.jit` — **the framework applies it for you**.

SUMMARY

# What has to survive this chapter

1. **Progressive disclosure of complexity** — one spectrum of workflows over the shared `Layer` and `Model` APIs, not several frameworks.
2. **Sequential** for simple stacks; **Functional API** for graphs of layers, and that is what the rest of the book uses; **subclassing** for anything that is not a graph.
3. Functional gives **access to layer connectivity**: `plot_model()` and feature extraction. Subclassing **forfeits both**.
4. The best combination in practice: **a Functional model containing subclassed layers**.
5. **Custom metrics** = `update_state` / `result` / `reset_state`. **Callbacks** = the six `on_*` hooks.
6. **`EarlyStopping`** + **`ModelCheckpoint`** replaces the wasteful train-twice pattern.
7. Writing `train_step()`: remember **`training=True`** and **`trainable_weights`**; in PyTorch **`zero_grad()`**; in JAX everything goes through the **`stateless_*`** family.

NOTEBOOK

03_custom_train_step_per_backend.ipynb

NEXT

Chapter 8 — Image classification